

Question Bank – Unit 3

Part A – Short Answer Questions (2 Marks)

1. Define concurrency in an operating system.
2. What is interleaving in concurrent execution?
3. Define mutual exclusion.
4. What is meant by a race condition?
5. Define process synchronization.
6. What is a semaphore?
7. What is a Resource Allocation Graph (RAG)?
8. Define deadlock in operating systems.
9. List the four necessary conditions of deadlock (Coffman conditions).
10. What is starvation in operating systems?

Part B – Long Answer Questions (10 Marks)

1. Explain the concept of concurrency in operating systems. Discuss the principles and advantages of concurrency.
2. Explain process synchronization and its significance. Describe the synchronization mechanisms such as mutual exclusion, semaphores, monitors, and condition variables.
3. Explain the Producer–Consumer problem with the help of semaphores.
4. What is deadlock? Explain the four necessary conditions for deadlock with suitable examples.
5. Explain the Dining Philosophers problem and discuss possible solutions to avoid deadlock and starvation.